Name: _____ Period: _____

IN CLASS LAB 1

The Lake Nyos Mystery

BACKGROUND
On August 21, 1986, a disaster struck Lake Nyos in Cameroon, Africa. A cloud swept through nearby villages and killed nearly 1,700 people and thousands of livestock in a single night. Survivors described strange details: oil lamps going out, a warm feeling in their bodies, and unconsciousness that lasted for hours. Buildings were untouched. Plants were untouched.

How could so many people and animals suddenly die without any signs of destruction?

What you are doing this week

Day	The question we are answering	What you will have when you finish
1	Where and what actually happened?	Coordinates of three lakes, an evidence chart, and a comparison
2	How could a lake hold that much gas?	Four demo observations and your first idea
3	How did the gas move, and why so sudden?	A group investigation, a model of Earth's systems, and a feedback loop
4	What do we do about it?	Your first CER and a solution you can defend

LEARNING INTENTION
Today we will be learning how matter moves between Earth's systems so we can explain a real disaster using evidence instead of guessing.

SUCCESS CRITERIA
- I can record what I observe without jumping to an explanation.
- I can locate a place using latitude and longitude and say why its location matters.
- I can trace how matter moved from one Earth system to another and explain what caused each move.
- I can write a claim, back it with evidence, and explain why that evidence proves it.

Two things to know before we start. First, you do not need to already know any of this. Everything you need is either in this packet or in something you will see in class. Second, nobody knew what happened at Lake Nyos for years, and scientists argued about it. So if you do not have the answer on Day 1, you are exactly where the real scientists were.

DAY 1: What Actually Happened?

Your job today: collect evidence. Do not explain anything yet. Write what was observed, not what you think caused it.

> **Observation vs. explanation.** An **observation** is something you can see, hear, measure, or read in a report. "The lamps went out" is an observation. An **explanation** is your idea about why. "There was no power to the lamp" is an explanation. Today, only observations.

TASK 1: Where in the World Did This Happen?

Before we ask what happened, we have to know where. Scientists locate any place on Earth with two numbers:

Latitude	How far **north or south** of the equator. The equator is 0°. The poles are 90°N and 90°S. Lines run east to west.
Longitude	How far **east or west** of the Prime Meridian, which is 0° and runs through Greenwich, England. Lines run north to south.

Two rules that matter: latitude is always written first, and the direction letter is never optional. 6°N and 6°S are almost 830 miles apart.

Lake	Latitude	Longitude	Country
Lake Nyos	6° 26′ N	10° 18′ E	Cameroon
Lake Monoun	5° 35′ N	10° 35′ E	Cameroon
Lake Kivu	2° 00′ S	29° 10′ E	Rwanda / DR Congo

Directions: Use the coordinates above as well as the map provided in your Cameroon Gazette packet.		
1	Which lake is in the Southern Hemisphere? How do you know?	
2	Which two lakes are closest together? Which two numbers told you?	
3	Which lake sits closest to the equator?	
4	How many degrees of latitude separate Lake Nyos and Lake Kivu? Careful, they are in different hemispheres.	

5. All three of these lakes sit in a region with volcanic activity underground. Why might that matter? Write an assumption. (You are not expected to have the correct answer yet.)

TASK 2: Observations from the Articles
Read the article excerpts and photos. Fill in each row with what people saw, heard, felt, or measured.

Category	Evidence / What happened	Notes and details
Survivor accounts		
Lake conditions		
Environmental effects		
Scientific investigations		

TASK 3: Two Lakes, Two Years Apart
You just located Lake Monoun. Something similar happened there in 1984, two years before Nyos. Use the observation lists in the Evidence Packet

Feature	Lake Monoun, 1984	Lake Nyos, 1986
Setting/type of lake		
Deepest point		
Sounds or movement reported		
Who or what died		
What was <u>not</u> damaged		

1. List three things these two events have in common.

1.

2.

3.

2. Which single piece of evidence is the strangest to you, and why?

DAY 1 EXIT

Write one question you would need answered before you could explain what happened here.

DAY 2: How Could a Lake Hold That Much Gas?

Your job today: watch each demo and figure out which part of the mystery it explains.

TASK 1: Demo Observations

Write what you saw first. Then write what it might explain about the lake. Use your observation sheet and Venn diagram alongside this page.

Demo	What I observed	What this could explain about Lake Nyos
1. Density tower		
2. Hot water and cold water		
3. Cold ice cubes in warm water		
4. Warm soda vs. cold soda		

TASK 2: Patterns You Notice

1. Demos 1, 2, and 3 all show the same idea. State that idea in one sentence.

_____

_____

2. Answer these about the soda demo:

A. When both cans were opened, which one lost its gas faster, the warm can or the cold can?

_____

B. Lake Nyos has warm water at the surface and cold water at the bottom. Based on your answer to part a, which layer can hold more dissolved gas?

_____

C. The gas stayed dissolved in the can until something changed. What changed?

_____

3. Make a connection.
Lakes in western New York freeze in winter and warm up in summer, and the water layers flip over and mix completely twice a year. Lake Nyos sits near the equator, where the surface water stays warm all year and the layers almost never mix.

Use today's demos to explain why a lake near Buffalo is unlikely to build up dissolved gasses, even if there were volcanic vents at the bottom.

Three terms you will need tomorrow:

Term	Definition	You saw this in
Density	How much mass is packed into a given amount of space. Denser material sinks below less dense material.	Demo 1
Stratification	When water separates into layers that do not mix because each layer has a different density.	Demos 2 and 3
Solubility	How much gas a liquid can hold. Cold liquids hold more dissolved gas than warm ones.	Demo 4

DAY 2 REFLECTION: Your First Idea
What do you think caused the Lake Nyos disaster? Back it up with at least two pieces of evidence from Day 1 or Day 2.

This is a *first* idea, not a final answer. You will rewrite it properly on Day 4.

What you still cannot explain
Today's demos explain how the gas got into the lake and stayed there. They do not explain two things:

- Why did people and cattle die as far as six miles away, while the plants and buildings right next to them were untouched?
- Why did it all happen in one night instead of leaking out slowly over years?

Pick the question you think is harder to answer. Explain what evidence you would need in order to answer it.

DAY 3: How Did the Gas Move, and Why So Sudden?

Your job today: answer the two questions you were left with yesterday.

TASK 1: The Milk Jug Investigation

Work in your group. You will pour something invisible out of a jug and over a row of lit candles.

> **Safety.** Tie back long hair and push sleeves up before lighting anything. One person handles the matches. Tell your teacher right away if a candle tips over.

Directions:

1	Set the candles in the clear bin, spaced a few inches apart. Do not crowd them against the wall of the bin.
2	Write your prediction in the box below. Do this before you light anything.
My prediction, written before we started	
3	Light the candles.
4	Add the baking soda to the jug, then pour in the vinegar. Swirl 2-3x and leave the jug open.
5	Wait about 10-15 seconds while the jug fills.
6	Tilt the jug slowly over the bin as if pouring water, keeping the mouth of the jug in the corner that has the highest candle. Do not let any liquid spill into the bin.
7	Record what you observe.
What actually happened	

1. Carbon dioxide is invisible, so you could not see it pour. What did you see instead that told you it was moving?

2. Gas rising away into the sky is what most people expect. This gas went down instead. What does that tell you about carbon dioxide compared to air?

3. Now answer yesterday's question. Why did people and cattle die as far as six miles away, while the plants and buildings next to them were untouched?

TASK 2: Sort Your Evidence Into Earth's Systems

Fill in the second column using your own evidence from Days 1, 2, and 3.

System	At Lake Nyos, this includes...
Geosphere	
Hydrosphere	
Atmosphere	
Biosphere	

These four systems are not separate. Matter and energy move between them constantly. That movement is what this whole course is about, and Lake Nyos is one very dramatic example.

TASK 3: Build the Model

Each row asks one question. Write your answer in the box on the right. The shaded rows are the four systems. The arrow rows ask what caused the gas to move from one system to the next.

	The question to answer	Write your answer here
STEP 1: GEOSPHERE ↓	Where did the carbon dioxide come from in the first place? (What is underneath a volcanic crater lake?)	
geosphere to hydrosphere	What moved it into the water?	
STEP 2: HYDROSPHERE ↓	Where did the gas get stored, and why did it stay there for years? (Use two of your Day 2 words.)	
hydrosphere to atmosphere	What disturbed the lake in 1986?	
STEP 3: ATMOSPHERE ↓	What happened to the gas once the deep water was disturbed?	
atmosphere to biosphere	Why did the gas stay in the valley instead of rising up?	
STEP 4: BIOSPHERE	What did the gas do to living things, and why were plants and buildings unharmed?	

TASK 4: Match the Demo to the Explanation
Write the letter of the density explanation each demo shows. Letters may be used more than once. **One demo does not show density at all.** For that one, write "not density" and name what it does show.

Density explanations:
A. Why the cold deep water stayed at the bottom and never mixed with the warm water above it
B. Why the released gas cloud sank into the valleys instead of rising into the sky
C. This demo does not show density

Demo	Letter	Demo	Letter
1. Density tower (no movement)		4. Warm soda vs. cold soda	
2. Hot water over cold water (no movement)		5. Milk jug over candles	
3. Cold ice cubes in warm water with "wind"		*Density does one more job in this story that no demo showed you. You will find it in Task 5.*	

TASK 5: Why Was It Sudden?
Your model explains what moved. It does not yet explain why it happened all at once instead of slowly leaking out over years. For that you need one more idea.

Feedback loops
A feedback loop is when a change causes an effect, and that effect comes back around and changes the original thing again.

Positive feedback loop	The change feeds itself and speeds up. Small starts become big fast. Also called a reinforcing loop.
Negative feedback loop	The change works against itself and settles down. Also called a balancing loop.

Careful. Positive does not mean good and negative does not mean bad. They mean the change is speeding up or slowing down. The Lake Nyos loop is positive, and it killed 1,700 people.

Fill in the loop below using what you already know about density and solubility.

START	Something disturbs the lake and deep water starts moving upward.
↓	**As the water rises, the pressure on it drops. What does that do to the dissolved gas?** Answer:_____
↓	**CO_2 bubbles are released in the water. What does that do to the water's density?** Answer:_____
↓	**The water is now less dense than it was. What does less dense water do, and what happens to the pressure on it as it moves?** Answer:_____
LOOP	**↑ Back to the top. The whole thing repeats, faster each time.**

1. Is this a positive or a negative feedback loop? How do you know?

2. Use your loop to explain the speed of the event.

   A. Go around the loop one time. At the end of that trip, is more gas coming out of solution than at the start, or less?

   B. Each trip around the loop makes the next trip happen faster. Which step in the loop is responsible for that?

   C. Now put it together. In two or three sentences, explain why the gas came out in one night instead of leaking out slowly over many years. Use at least two steps from your loop in your answer.

DAY 3 EXIT

Which step in your model or your loop were you least sure about? What would help you be sure?

DAY 4: What Do We Do About It?

Your job today: judge the solutions, then write your explanation.

CER: how scientists write an explanation

Every explanation in this class has three parts. You will use this all year.

CLAIM	Your answer to the question. One clear sentence. No maybe.
EVIDENCE	Specific facts, observations, or measurements. Not your opinion. Something someone else could check.
REASONING	Why that evidence proves your claim. This is the part people skip, and it is the part that matters most.

TASK 1: Evaluate the Solutions

Engineers were asked to make Lake Nyos safe. A solution has to work scientifically **and** be possible for the people who actually live there.

The situation: the villages near Lake Nyos are rural. Electricity is limited, and fuel is expensive and hard to get. A solution that only works while somebody keeps paying to run it is not really a solution for this place.

Proposed solution	How would this reduce the danger?	What would people need to pay for or supply every year to keep it working?	Is that realistic near Lake Nyos? Why or why not?
Move the villages to higher ground			
Powered pumps that pull deep water to the surface			
Plant trees around the lake			
Self-sustaining degassing pipes**			

****How a self-sustaining degassing pipe works.** A pipe runs from the deep water up to the surface. A pump gives it one push to get water moving. After that, the pressure drops, dissolved gas comes out of solution, the bubbles make the water less dense, and it rises on its own, pulling more water up behind it. The pump can be switched off. **Self-sustaining** means it keeps running on its own once it starts

Turn back to Day 3, Task 5. That is the loop you filled in showing why the gas came out all at once.

1	Compare the self-sustaining degassing pipe description above to the steps in that loop. Write down two steps that appear in both.
2	In 1986 this loop happened by accident, all at once, and people died. In the pipe, engineers start the same loop on purpose. Explain what makes the pipe version safe when the 1986 version was not.
3	Over many years the pipes remove gas from the lake, so the chance of a sudden release keeps dropping. Is that a positive or a negative feedback loop?

Which solution is best? Name the real-world limit that decides it, not just the science.

☐ **Check yourself:** more than one of these removes the carbon dioxide. So "it removes the CO_2" cannot be what makes one of them the best. What separates them?

TASK 2: Your CER

Question: What caused the Lake Nyos disaster, and what was done to prevent it from happening again?

CLAIM: one clear sentence stating the cause.

EVIDENCE: three specific pieces from the articles, the demos, or your Day 3 model.

1	
2	

3	

REASONING:

1. Which of these explains your evidence: density, stratification, solubility, or the feedback loop?
2. Explain how that makes your evidence matter. *Example: because CO_2 is denser than air, the cloud spread into valleys instead of rising away.*
3. Rule something out. Why was this not a fire, an explosion, or a flood?

SOLUTION: Name the solution you would choose. Then explain what makes it better than the other one that also removes the carbon dioxide.

Before you turn this in:

☐ My claim answers the question that was asked ☐ Three separate pieces of evidence
☐ My reasoning explains why the evidence matters, not just "because of the gas" ☐ My reasoning rules something out
☐ ☐ My solution explains why it beats the other one that also removes the gas it

DAY 4 EXIT

Lake Kivu *sits on the border of Rwanda and the Democratic Republic of the Congo. It is a deep lake in a volcanic region, and about two million people live along its shores. Scientists have measured large amounts of carbon dioxide dissolved in its deep water.*

1. Explain how carbon dioxide from the geosphere could end up stored in the deep water of a lake like this.

2. Explain what could make that gas move into the atmosphere, and why the release would happen quickly rather than slowly.